

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Problem Set 13

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Topic

Dense Media and the Collapse Limit

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Dense Media and the Collapse Limit. State all assumptions and conventions.

$$dP/dr = -\rho d\Phi/dr + f_m(r; \theta_D)$$

2. Evaluate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$\omega_0^2(M_c) = 0$$

3. Compare the third relation in two limiting regimes and explain the physical meaning of the result.

$$M_c = 1.9M_\odot$$

Show enough intermediate work that another student can reproduce the calculation.